Luc's Statistical Coffee Price Forecast

Monthly Model Report

2026-07-18

Executive Summary

Coffee Price Forecast Summary

The model forecasts a declining trend for both arabica and robusta over the next three months. Arabica is expected to decline from the current 307.8 *¢/lb* to 275.9 *¢/lb* by month three, while robusta drops from 169.4 *¢/lb* to 160.4 *¢/lb*. Forecast uncertainty widens modestly with longer horizons. For arabica, the 80% forecast range spans about 51.7 *¢/lb* at one month, narrowing to 48.4 *¢/lb* by month three. Robusta shows tighter bands, ranging from 27.7 *¢/lb* to 26.9 *¢/lb* across the same period.

Historically, the model has demonstrated solid near-term accuracy. One-month forecasts achieved mean errors of 6.1% for arabica and 5.6% for robusta. Accuracy declines at longer horizons, with three-month errors reaching 11.4% and 11.9% respectively. Coverage data on actual price containment are currently unavailable.

The spread signal between arabica and robusta sits at a z-score of -0.14, indicating a neutral stance. The signal suggests the arabica-robusta premium is neither unusually wide nor tight, with an estimated mean reversion half-life of approximately 23 months.

Current Forecasts

Median (p50) price forecasts and 80% confidence intervals (p10–p90) for the next three months.

Symbol	Horizon	Median (¢/lb)	80% CI (¢/lb)	Band width
Arabica (KC=F)	h=1	294.5	[272.9, 324.6]	51.7
Arabica (KC=F)	h=2	283.5	[262.8, 312.6]	49.8
Arabica (KC=F)	h=3	275.9	[255.7, 304.1]	48.4
Robusta (RM=F)	h=1	165.3	[152.0, 179.7]	27.7
Robusta (RM=F)	h=2	162.3	[149.3, 176.5]	27.2
Robusta (RM=F)	h=3	160.4	[147.6, 174.5]	26.9

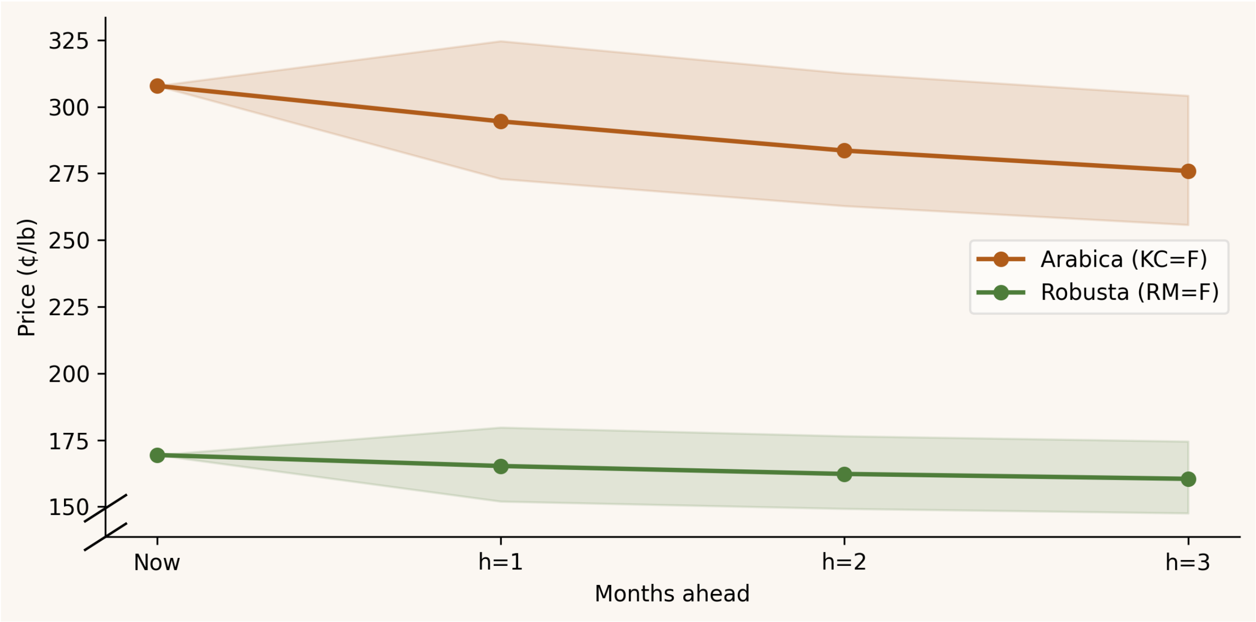

Figure 1: Price trajectory with 80% confidence ribbon

Model Performance

Mean absolute error (MAE), mean absolute percentage error (MAPE), and 80% interval coverage across all walk-forward backtest windows.

Symbol	Horizon	MAE (¢/lb)	MAPE	Coverage 80%
Arabica	h=1	12.8	6.1%	N/A
Arabica	h=2	19.8	9.2%	N/A
Arabica	h=3	25.3	11.4%	N/A
Robusta	h=1	7	5.6%	N/A
Robusta	h=2	11.3	9.1%	N/A
Robusta	h=3	15.1	11.9%	N/A

Coverage 80% target: 80% of forecast ranges should contain the realized price. Values below 80% indicate the model is over-confident.

Recent Forecast Performance

One-month-ahead forecasts versus realized prices over the last three months.

Month	Symbol	Forecast (¢/lb)	Actual (¢/lb)	Error
2026-03	Arabica (KC=F)	312.2	334.1	-6.6% (below actual)
2026-03	Robusta (RM=F)	170.9	176.8	-3.3% (below actual)
2026-02	Arabica (KC=F)	360.8	321.4	+12.3% (above actual)
2026-02	Robusta (RM=F)	192.2	179.6	+7.0% (above actual)
2026-01	Arabica (KC=F)	371.1	363.9	+2.0% (above actual)
2026-01	Robusta (RM=F)	182.2	192.3	-5.2% (below actual)

January 2026

The model forecast Arabica at 371.1 *¢/lb* and Robusta at 182.2 *¢/lb*, with actual prices settling at 363.9 *¢/lb* and 192.3 *¢/lb* respectively. Arabica missed by just 2.0 percent on the high side, while Robusta underperformed the forecast by 5.2 percent. The Robusta miss likely reflects unexpected Brazilian currency strength or a shift in Indonesian supply expectations that the historical relationships failed to capture.

February 2026

The model overestimated both contracts significantly, projecting Arabica at 360.8 *¢/lb* and Robusta at 192.2 *¢/lb* against actuals of 321.4 *¢/lb* and 179.6 *¢/lb*, translating to misses of 12.3 percent and 7.0 percent respectively. This larger divergence suggests the model did not anticipate a meaningful demand softening or a shift in macroeconomic sentiment that pressured both varieties lower. A potential shift in fund positioning or emerging concerns about global growth would not be well-captured by historical price correlations alone.

March 2026

The forecast reversed course with Arabica predicted at 312.2 *¢/lb* and Robusta at 170.9 *¢/lb*, but actual prices rallied to 334.1 *¢/lb* and 176.8 *¢/lb*, missing by 6.6 percent and 3.3 percent respectively. The rally likely stemmed from a supply-side development such as adverse weather in a major producing region or reduced output expectations that a backward-looking model could not foresee. Near-term fund covering or a shift in physical buying patterns would similarly fall outside the scope of purely statistical forecasting.

Arabica–Robusta Spread Signal

The spread model tracks whether Arabica is unusually cheap or expensive relative to Robusta. It measures the **z-score** of the log price ratio — how many standard deviations the current ratio sits from its long-run average. A z-score near zero means the relationship is normal; a z-score beyond ± 2 means a rare dislocation (this happens roughly 5% of the time historically) that has tended to revert. A trade signal fires at $|\mathbf{z}| > \mathbf{2}$ and closes once the ratio normalises back to $|\mathbf{z}| < \mathbf{0.5}$.

As of	2026-06-01
Current signal	**Neutral** — ratio within normal range, no edge detected
Z-score	-0.14 std devs from average
Half-life	22.8 months — typical time to revert to average
Entry threshold	
Exit threshold	

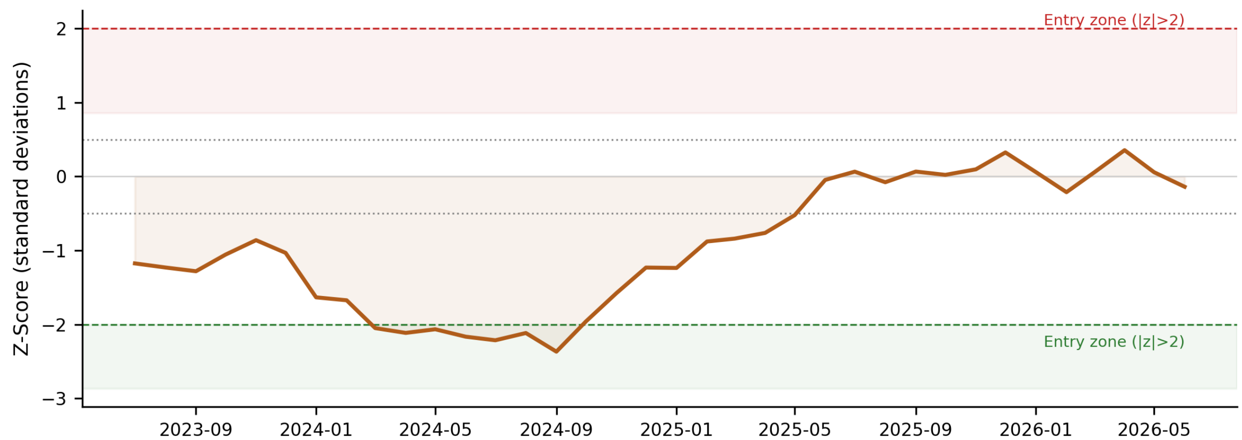

Figure 2: Z-score of Arabica/Robusta log spread — last 36 months. Red zone = Arabica expensive; green zone = Arabica cheap. Dashed lines mark entry thresholds (± 2 std devs); dotted lines mark exit thresholds (± 0.5).

Data & Methodology

Data sources: Arabica (KC=F) and Robusta (RM=F) monthly prices from FRED (`PCOFFOTMUSDM`, `PCOFFROBUSDM`). FX drivers (BRL/USD, VND/USD, IDR/USD) from Alpha Vantage. DXY broad index from FRED (`DTWEXBGS`). Data starts 2000-01-01; monthly series is the mean of daily adjusted closes within each calendar month.

VECM: A two-variable Vector Error Correction Model is fit on log Arabica and log Robusta prices, with BRL, VND, IDR, and DXY as exogenous drivers. Cointegration rank is fixed at 1 (confirmed by Engle-Granger test, p=0.009). Lag order is selected by AIC each run. Point forecasts at 1, 2, and 3 months use a naïve FX assumption (hold FX at last observed value).

GAMLSS SHASH: A Sinh-ArcSinh distribution is fit (via R's `gamlss` package) to the VECM forecast residuals, separately per symbol and per volatility regime (Low / Medium / High, defined by rolling 12-month standard deviation). This gives a full predictive distribution. Final price quantiles are `price_qX = vecm_point × exp(gamlss_residual_qX)`.

Spread signal: An AR(1) model is fit to the log spread series; the expanding z-score triggers entry at $|z| > 2$ and exit at $|z| < 0.5$.

Report generated by the `coffee-forecast` pipeline.